

Unitary or Reversible X or NOT Gate Controlled NOT or C-NOT Gate

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Unitary Matrix

From Dancing With Qubits: Chapter 5 - page 184

5.7.5 Unitary transformations

What are the characteristics of linear transformations that preserve length? That is, if $L : V \rightarrow V$ and it is always true that $\|L(\mathbf{v})\| = \|\mathbf{v}\|$, what can we say about the matrix of L ?

If U is a complex square matrix then it is *unitary* if its adjoint $U^\dagger$ is also its inverse U^{-1} . Hence $UU^\dagger = U^\dagger U = I$. This means the columns of U are orthonormal, as are the rows.

We call this symbol “Dagger” or the Adjoint

The matrices corresponding to gates/operations on qubits are all unitary.

Classical Boolean logic

- Represented as a truth table.

	x_0	x_1	$x_0 \wedge x_1$
	0	0	0
AND gate:	0	1	0
	1	0	0
	1	1	1

	x_0	$\neg x_0$
	0	1
NOT gate:	1	0

OR is simulatable by
AND and NOT:

$$a \vee b = \neg(\neg a \wedge \neg b)$$

	x_0	x_1	$x_0 \vee x_1$
	0	0	0
OR gate:	0	1	1
	1	0	1
	1	1	1

NAND is universal for
deterministic circuits

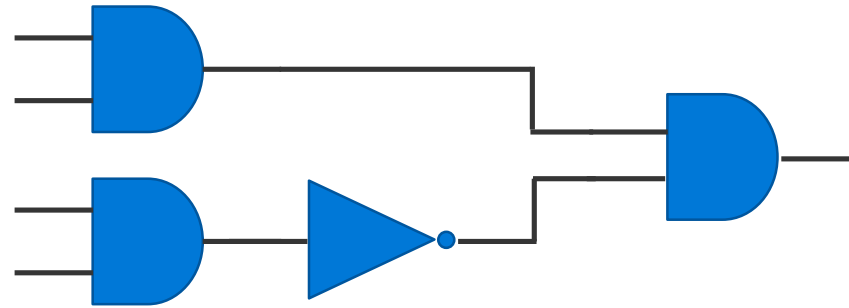
	x_0	x_1	$\neg(x_0 \wedge x_1)$
	0	0	1
NAND gate:	0	1	1
	1	0	1
	1	1	0

Classical Boolean circuits

- Universal gate set: set of elementary gates from which any circuit can be built.



- We can compose gates to make larger circuits.



- *Challenge*: Quantum computation requires reversible gates. Many gates are not reversible.

Quick review: Why must quantum gates be reversible?

- A reversible gate has a *unique* input state for every output state.
- Quantum mechanics dictates that the evolution of (computation on) a quantum state must be reversible.
- Mathematically, quantum gates can be represented as matrices and quantum states as vectors.
- The matrix must be unitary
$$UU^\dagger = U^\dagger U = I$$

†: Adjoint, aka
complex conjugate
transpose, aka
inverse

Classical Boolean logic

Are any of these gates reversible?

	x_0	x_1	$x_0 \wedge x_1$
	0	0	0
AND gate:	0	1	0
	1	0	0
	1	1	1

	x_0	x_1	$x_0 \vee x_1$
	0	0	0
OR gate:	0	1	1
	1	0	1
	1	1	0

	x_0	$\neg x_0$
NOT gate:	0	1
	1	0

	x_0	x_1	$\neg(x_0 \vee x_1)$
	0	0	1
NAND gate:	0	1	1
	1	0	1
	1	1	0

Classical Boolean logic

Let X represent the NOT gate and b represent the state of the bit. We can represent X as a matrix acting on a bit vector:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$
$$X|b\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b \\ 1-b \end{pmatrix} = \begin{pmatrix} 1-b \\ b \end{pmatrix}$$

Applying X twice returns the original result:

$$X(X|b\rangle) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b \\ 1-b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b \\ 1-b \end{pmatrix}.$$
$$X^2 = \mathbf{1}.$$

This is the Identity gate, denoted as Id , I , or $\mathbf{1}$.

Classical Boolean logic

- What happens if we apply the gate to a quantum state?
- We permute the amplitudes!

$$X|\psi\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_0 \end{pmatrix}$$
$$X|\psi\rangle = X(c_0|0\rangle + c_1|1\rangle) = c_0|1\rangle + c_1|0\rangle$$

- Also called the "Pauli X" gate

$$|a\rangle \rightarrow |\neg a\rangle$$

Verify X is unitary.

Classical Boolean logic

What about the XOR gate?

XOR gate:	x_0	x_1	$x_0 \oplus x_1$	
	0	0	0	00 → 00
	0	1	1	01 → 01
	1	0	1	10 → 11
	1	1	0	11 → 10

Note that keeping one of the input bits as output yields a unique output.

You could interpret this as flipping (applying X) to the second bit when the first bit is 1.

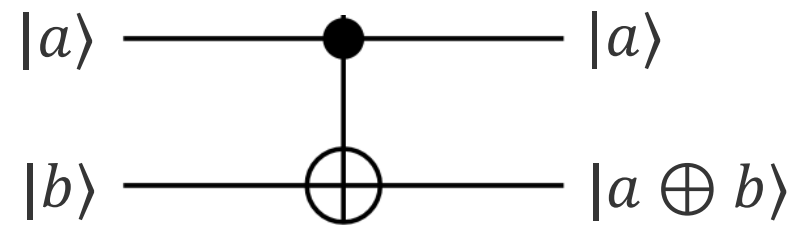
Controlled-NOT gate

$$|a, b\rangle \rightarrow |a, a \oplus b\rangle$$

If a is 1, then b is flipped
 $\Rightarrow b$ undergoes an X gate

$$CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

00 $\rightarrow$ 00
01 $\rightarrow$ 01
10 $\rightarrow$ 11
11 $\rightarrow$ 10



Controlled-NOT gate

$$|a, b\rangle \rightarrow |a, a \oplus b\rangle$$

If a equals 1, then b is flipped
 $\Rightarrow b$ undergoes an X gate

Verify CX is unitary.

$$CX|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} c_0 \\ c_1 \\ c_3 \\ c_2 \end{pmatrix}$$

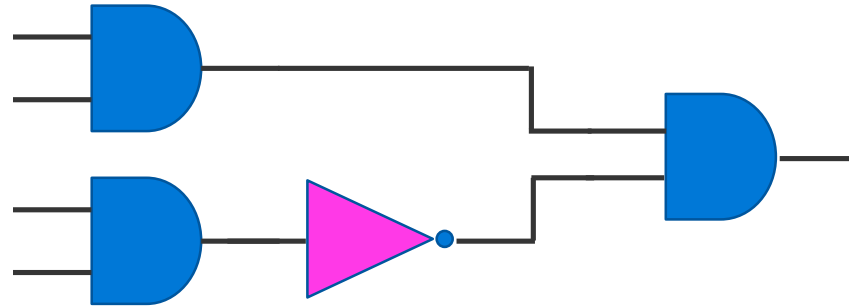
Example:

$$CX|+, 0\rangle = CX \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

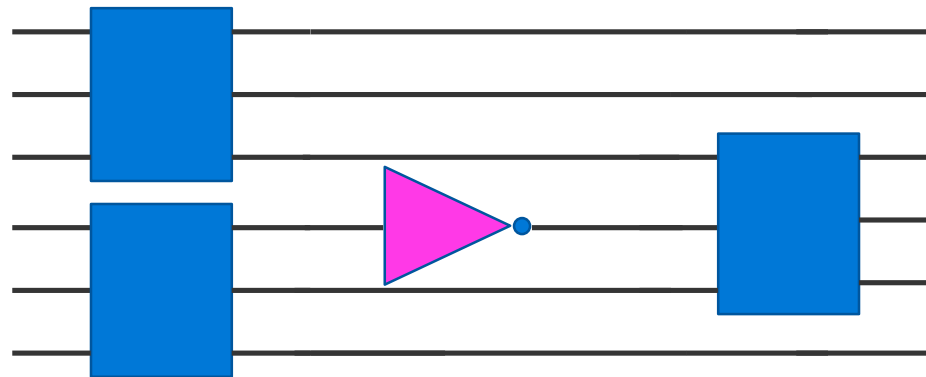
Verify this using matrix – vector multiplication.

Reversing a circuit

Example:



Replace each gate with a reversible one:



Quick review: circuit diagrams

- Circuits are read like music: left to right
- Wires (staff lines) represent information (bit or qubit)
- Boxes and symbols represent operations
- Note that circuit diagrams read left to right, and are mathematically applied right to left:

$$C_m C_{m-1} \dots C_0 |\psi\rangle$$

